

UNIVERSITY OF BRISTOL
MSc BIOINFORMATICS

**Evaluating Evapotranspiration Generalisation Under a
Changing Climate using Machine Learning**

PROJECT PLAN

Author

Archie BENN

Supervisor

Professor Martin DE KAUWE

ABSTRACT (The abstract has a maximum of 200 words)

Key words: Machine Learning, FLUXNET, Evapotranspiration.

Word count: 3000

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie BENN

April 30, 2026

CONTENTS

Literature Review (3000 words)	4
Evapotranspiration and The Water Cycle	
<i>(Why does the water cycle matter?)</i>	4
Evapotranspiration	4
Eddy Covariance	4
The Changing Climate	
<i>(How is the water cycle changing?)</i>	4
Evapotranspiration Prediction	
<i>(What don't we know and what are the issues?)</i>	4
Contemporary ET Prediction Methods	4
Machine Learning Models for ET Prediction	4
Project Scope and Machine Learning Applicability	
<i>(What is the case for using ML here?)</i>	4
Project Scope	4
Why is ML applicable here?	4
Project Aims and Objectives (500 words)	5
Aims	5
Project Plan (500 words)	6
Data Acquisition and Selection	6
Analysis	6
Feature Engineering	6
ML Model Setup and Training	6
ML Model Evaluation	6
ML Model Predictions	6
Tools	6
Interpretation	6
Gantt Chart	6
Risk Register (~ 250 words)	7
Figures and Tables	8

LITERATURE REVIEW (3000 WORDS)

EVAPOTRANSPIRATION AND THE WATER CYCLE

(Why does the water cycle matter?)

Evapotranspiration

Eddy Covariance

THE CHANGING CLIMATE

(How is the water cycle changing?)

EVAPOTRANSPIRATION PREDICTION

(What don't we know and what are the issues?)

Contemporary ET Prediction Methods

Machine Learning Models for ET Prediction

TABLE

PROJECT SCOPE AND MACHINE LEARNING APPLICABILITY

(What is the case for using ML here?)

Project Scope

Why is ML applicable here?

PROJECT AIMS AND OBJECTIVES (500 WORDS)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

PROJECT PLAN (500 WORDS)

DATA ACQUISITION AND SELECTION

ANALYSIS

Feature Engineering

ML Model Setup and Training

ML Model Evaluation

ML Model Predictions

TOOLS

INTERPRETATION

GANTT CHART

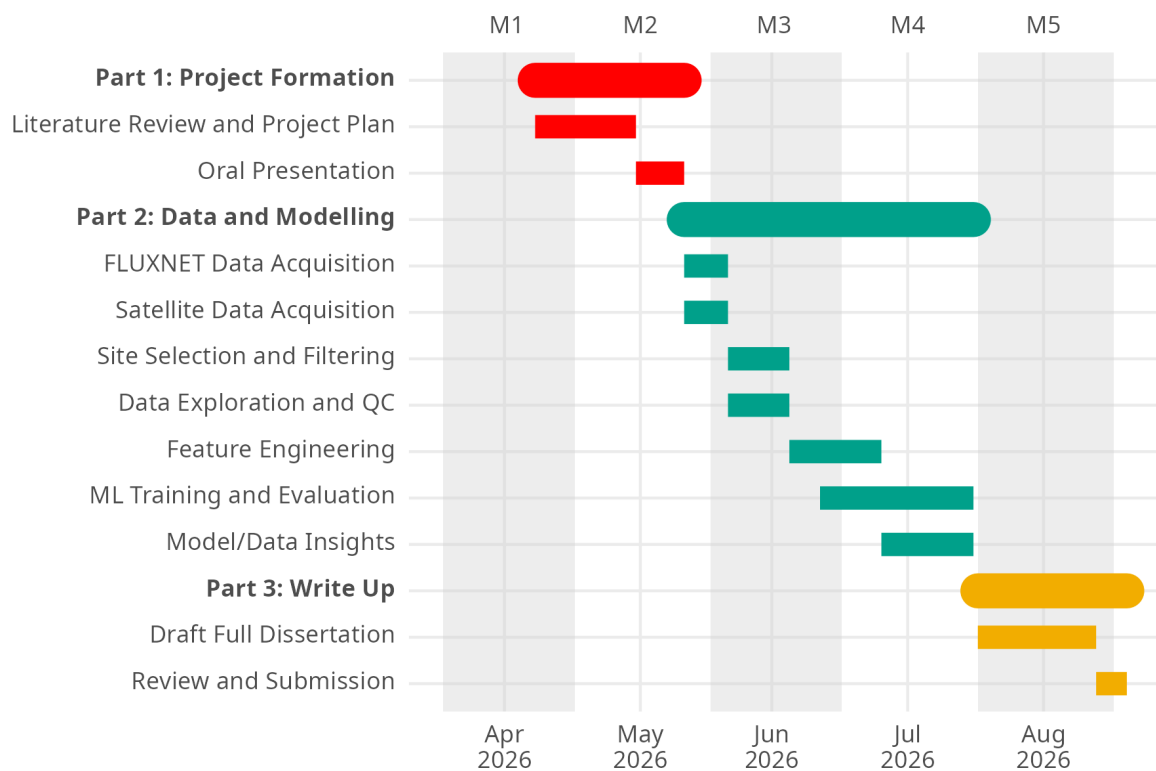


Figure 1: A proposed Gantt Chart for the duration of this project.

RISK REGISTER (~ 250 WORDS)

REFERENCES

FIGURES AND TABLES

Method	Advantages	Disadvantages
--------	------------	---------------